4.1.1 a) Given: triangle with vertex positions $p_1, p_2, p_3 \in \mathbb{R}^3$
and texture coordinates $t_i = \begin{pmatrix} u_i \\ v_i \end{pmatrix}$

$$\vec{e}_{12} = p_2 - p_1$$
$$\vec{e}_{13} = p_3 - p_1$$

$$\Delta u_{12} = u_2 - u_1 \qquad \Delta v_{12} = v_2 - v_1$$
$$\Delta u_{13} = u_3 - u_1 \qquad \Delta v_{13} = v_3 - v_1$$

since $p(u,v)$ is locally linear on a triangle,
a change in texture space $(\Delta u, \Delta v)$ causes 3D displacement

$$\Delta p = \Delta u \cdot \vec{p_u} + \Delta v \cdot \vec{p_v}$$

using edges from V_1:

$$\boxed{\begin{aligned} \vec{e}_{12} &= p_2 - p_1 = (u_2 - u_1) \cdot \vec{p_u} + (v_2 - v_1) \cdot \vec{p_v} \\ \vec{e}_{13} &\quad p_3 - p_1 = (u_3 - u_1) \cdot \vec{p_u} + (v_3 - v_1) \cdot \vec{p_v} \end{aligned}}$$

b) $J_T^p = \begin{pmatrix} \vec{p_u} & \vec{p_v} \end{pmatrix} \in \mathbb{R}^{3\times 2}$

$E = \begin{pmatrix} \vec{e}_{12} & \vec{e}_{13} \end{pmatrix} \in \mathbb{R}^{3\times 2}$

$\Delta T := \begin{pmatrix} \Delta u_{12} & \Delta u_{13} \\ \Delta v_{12} & \Delta v_{13} \end{pmatrix} \in \mathbb{R}^{2\times 2}$

$$E = J \cdot \Delta T$$

$$J = E\,(\Delta T)^{-1} \qquad \text{if } \Delta T \text{ invertable}$$

$$(\Delta T)^{-1} = \frac{1}{\det(\Delta T)} \begin{pmatrix} \Delta v_{13} & -\Delta u_{13} \\ -\Delta v_{12} & \Delta u_{12} \end{pmatrix}$$

$$\det(\Delta T) = \Delta u_{12}\,\Delta v_{13} - \Delta v_{12}\,\Delta u_{13}$$

$$(\Delta T)^{-1} = \frac{1}{\Delta u_{12}\,\Delta v_{13} - \Delta v_{12}\,\Delta u_{13}} \begin{pmatrix} \Delta v_{13} & -\Delta u_{13} \\ -\Delta v_{12} & \Delta u_{12} \end{pmatrix}$$

Common factor: $r = \dfrac{1}{\Delta u_{12}\,\Delta v_{13} - \Delta v_{12}\,\Delta u_{13}}$

$$\rightarrow \boxed{\begin{aligned} J_T^P &= \left(\vec{p_u} \quad \vec{p_v}\right) = r \left(\vec{e_{12}} \quad \vec{e_{13}}\right) \begin{pmatrix} \Delta v_{13} & -\Delta u_{13} \\ -\Delta v_{12} & \Delta u_{12} \end{pmatrix} \\ \vec{p_u} &= r\left(\vec{e_{12}}\, \Delta v_{13} - \vec{e_{13}}\, \Delta v_{12}\right) \\ \vec{p_v} &= r\left(-\vec{e_{12}}\, \Delta u_{13} + \vec{e_{13}}\, \Delta u_{12}\right) \end{aligned}}$$

4.1.2 a) Mipmaps fix texture aliasing, shimmering and moiré when a texture is minified (many texels map to one pixel) by using pre-filtered, lower-resolutions of a texture. During rendering, the sampler chooses an appropriate mip level based on the projected pixel footprint.

b) Given: square base img with resolution $r \times r$ (power of 2) and b bytes per pixel

$\rightsquigarrow r = 2^k$

mipchain stores in levels:

$r \cdot r,\ \frac{r}{2} \cdot \frac{r}{2},\ \frac{r}{4} \cdot \frac{r}{4}, \dots, 1 \cdot 1$ (ends on 1 pixel)

$\rightarrow$ pixel count is geometric series:

$r^2 + \left(\frac{r}{2}\right)^2 + \left(\frac{r}{4}\right)^2 + \dots + 1$ $\rightarrow$ first term $a = r^2$

$= r^2\left(1 + \frac{1}{4} + \frac{1}{16} + \dots + \frac{1}{r^2}\right)$ $\rightarrow q = \frac{1}{4}$ (half width & half height)

since $r = 2^k \Rightarrow k = \log_2(r)$

$\rightarrow$ last mip-level is $1 \Rightarrow$ Levels $i = 0, 1, 2, \dots, k$ (after k halvings, we reach 1)

$\Rightarrow k+1$ terms

Sum of geometric series:

For $a + aq + aq^2 + \dots + aq^k$

use $\sum_{i=0}^{k} a q^i = a \cdot \frac{1 - q^{k+1}}{1 - q}$.

With $a = r^2,\ q = \frac{1}{4}$:

$$P(r) = \sum_{i=0}^{k} r^2 \left(\frac{1}{4}\right)^i = r^2 \cdot \frac{1 - \left(\frac{1}{4}\right)^{k+1}}{1 - \frac{1}{4}} = r^2 \cdot \frac{4}{3}\left(1 - \left(\frac{1}{4}\right)^{k+1}\right)$$

$$\left(\frac{1}{4}\right)^k = \frac{1}{4^k} = \frac{1}{r^2}$$

$$\left(\frac{1}{4}\right)^k = \frac{1}{4^k} = \frac{1}{r^2}$$

$$\Rightarrow \left(\frac{1}{4}\right)^{k+1} = \left(\frac{1}{4}\right)^k \cdot \frac{1}{4} = \frac{1}{4r^2}$$

$$\Rightarrow P(r) = r^2 \cdot \frac{4}{3}\left(1 - \frac{1}{4r^2}\right) = \frac{4}{3}\left(r^2 - \frac{1}{4}\right)$$

$$\boxed{m_b(r) = b \cdot P(r) = b \cdot \frac{4}{3}\left(r^2 - \frac{1}{4}\right)}$$

$\rightarrow$ the "$-\frac{1}{4}$" term comes from the last level 1×1